

1. A voltage of **12 V** is applied to an electric motor with an armature resistance of **5.0 Ω** . When the armature is rotating at peak speed, the current in it is **0.48 A**. When a load is placed on the motor, its speed or frequency is reduced to **one-third of its peak speed**. What is the back EMF, **E_b** , of the motor under this load?
2. A town receives **10 MW** of power delivered at **50 kV** from a generator over transmission lines that have a resistance of **5.0 Ω** . What percentage of the power generated is lost as heat in the lines?